Device Control Panel (Firmware UI) — Feature List & State Machine Design

This document outlines the essential features and finite state machine (FSM) design for a device-side control panel using a 20x4 LCD and a 4x4 keypad. The design prioritizes simplicity, offline functionality, and quick access to critical operations.

1. Feature List

1.1 Core Functions

- Manual feeding with user-defined amount
- Quick preset feeding options (optional)
- View and toggle existing schedules
- Display real-time system status (feeder level, drinker level, power status)
- Display and acknowledge alerts (low levels, power failure)
- Basic settings (threshold adjustments, time setup)

1.2 UI Structure

- Home screen with summary information
- Menu-based navigation using keypad
- Dedicated pages for feeding, schedules, status, alerts, and settings
- Consistent navigation keys (A=Up, B=Down, C=Select, D=Back)

2. Finite State Machine (FSM)

2.1 State Definitions

- STATE_HOME
- STATE_MENU
- STATE_MANUAL_FEED_INPUT
- STATE_MANUAL_FEED_CONFIRM
- STATE_FEEDING_PROGRESS
- STATE_SCHEDULE_LIST
- STATE_SCHEDULE_TOGGLE
- STATE_STATUS_VIEW
- STATE_ALERTS_LIST

- STATE_ALERT_ACK
- STATE_SETTINGS_MENU
- STATE_SETTINGS_THRESHOLD
- STATE_SETTINGS_TIME

2.2 State Flow Overview

The system follows a hierarchical menu structure. Users navigate from the HOME screen to the MENU, then to specific feature states. Each state handles its own input and transitions.

```
enum State {
  STATE_HOME,
  STATE_MENU,
  STATE_MANUAL_FEED_INPUT,
  STATE_MANUAL_FEED_CONFIRM,
  STATE_FEEDING_PROGRESS,
  STATE_SCHEDULE_LIST,
  STATE_SCHEDULE_TOGGLE,
  STATE_STATUS_VIEW,
  STATE_ALERTS_LIST,
  STATE_ALERT_ACK,
  STATE_SETTINGS_MENU,
  STATE_SETTINGS_THRESHOLD,
  STATE_SETTINGS_TIME
};

void loop() {
  readKeypad();

  switch(currentState) {

    case STATE_HOME:
      drawHome();
      handleHomeInput();
      break;

    case STATE_MENU:
      drawMenu();
      handleMenuInput();
      break;

    case STATE_MANUAL_FEED_INPUT:
      drawFeedInput();
      handleFeedInput();
      break;

    // Additional states follow similarly...
  }
}
```

3. UI Screen Examples

Home Screen

```
Smart Feeder
Next: 08:00 AM
FeedLvl: 70%
[A]Menu   [C]Feed
```

Menu Screen

```
>Manual Feed
 Schedules
 Status
 Alerts
```

Manual Feed Input

```
Enter Amount:
[ 0.00 kg ]
C=OK  D=Back
```

Feeding Progress

```
Feeding...
Please wait
[====    ]
```

Schedule List

```
>08:00 ON
 12:00 OFF
 18:00 ON
C=Toggle D=Back
```

Status Screen

```
FeedLvl: 70%
Drink: 60%
Power: MAIN
D=Back
```

Alerts Screen

```
>Low Feed Level
 Power Failure
C=Ack  D=Back
```

Settings Menu

```
>Thresholds
 Time Setup
D=Back
```